

1.7 The current at the terminals of the element in Fig. 1.5 is

$$i = 0, \quad t = 0;$$

$$i = 20e^{-5000t} \text{ A}, \quad t \geq 0.$$

Calculate the total charge (in microcoulombs) entering the element at its left terminal.

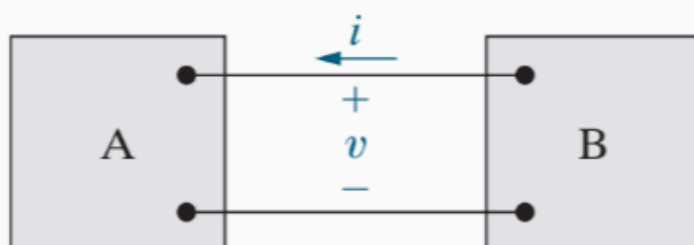
1.8 There is no charge at the left terminal of the element in Fig. 1.5 for $t < 0$. At $t = 0$ a current of $25e^{-1000t}$ mA enters the left terminal.

- Derive the expression for the charge that accumulates at the left terminal for $t > 0$.
- Find the total charge that accumulates at the left terminal.
- If the current is stopped at $t = 1$ ms, how much charge has accumulated at the left terminal?

1.15 Two electric circuits, represented by boxes A and B, are connected as shown in Fig. P1.15. The reference direction for the current i in the interconnection and the reference polarity for the voltage v across the interconnection are as shown in the figure. For each of the following sets of numerical values, calculate the power in the interconnection and state whether the power is flowing from A to B or vice versa.

- a) $i = 150 \text{ mA}$, $v = -20 \text{ V}$,
- b) $i = 15 \text{ A}$, $v = 40 \text{ V}$,
- c) $i = -50 \text{ mA}$, $v = -2 \text{ kV}$,
- d) $i = -3 \text{ A}$, $v = 80 \text{ V}$.

Figure P1.15



1.18 The voltage and current at the terminals of the circuit element in Fig. 1.5 are zero for $t < 0$. For $t \geq 0$ they are

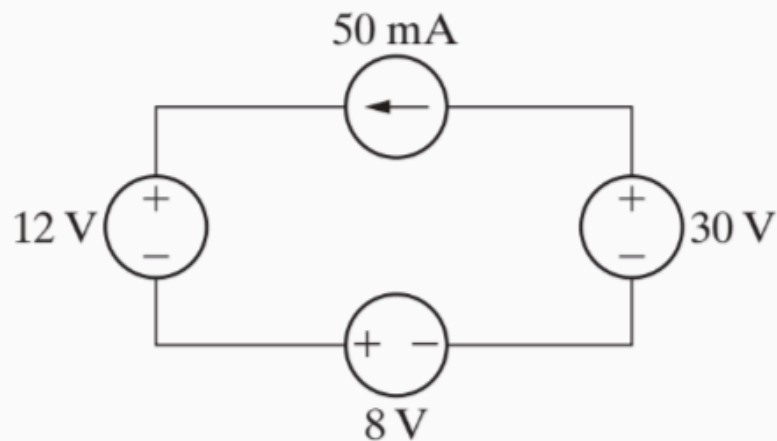
$$v = 80te^{-500t} \text{ kV},$$

$$i = 15te^{-500t} \text{ A}.$$

- a) Calculate the power supplied to the element at 10 ms.
- b) Calculate the total energy delivered to the circuit element.

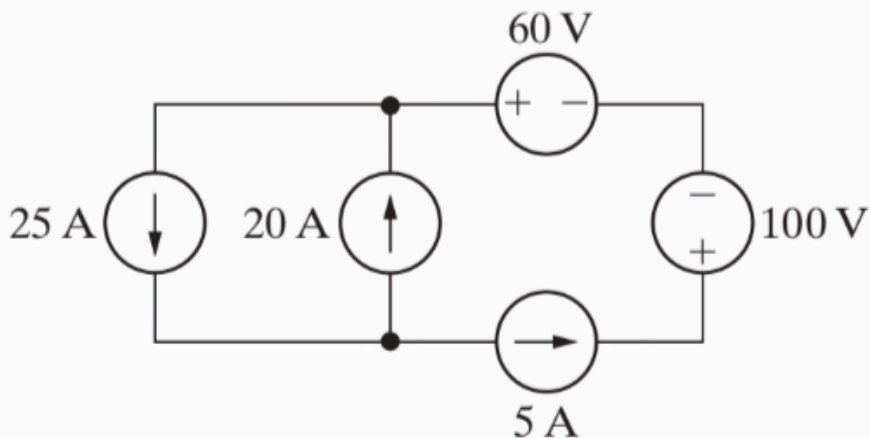
- 2.2** If the interconnection in Fig. P2.2 is valid, find the total power developed in the circuit. If the interconnection is not valid, explain why.

Figure P2.2



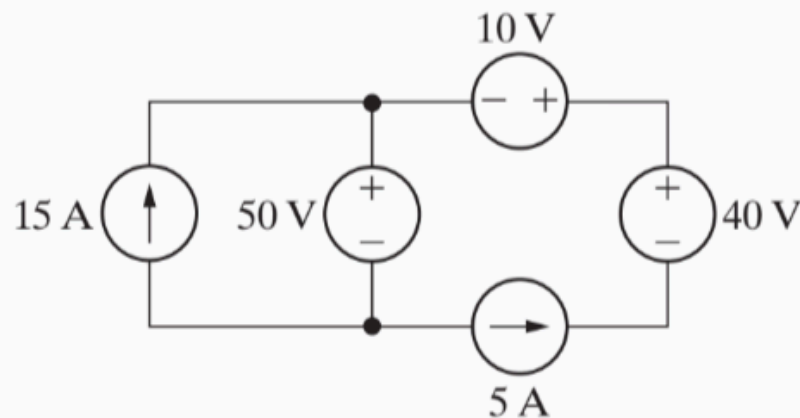
- 2.3** If the interconnection in Fig. P2.3 is valid, find the total power developed by the voltage sources. If the interconnection is not valid, explain why.

Figure P2.3



- 2.4** If the interconnection in Fig. P2.4 is valid, find the total power developed by the current sources. If the interconnection is not valid, explain why.

Figure P2.4



- 2.11** For the circuit shown in Fig. P2.11,
- Find i .
 - Find the power supplied by the voltage source.
 - Reverse the polarity of the voltage source and repeat parts (a) and (b).

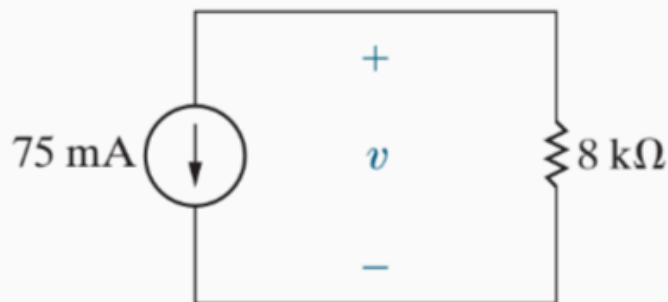
Figure P2.11



2.12 For the circuit shown in Fig. P2.12,

- Find v .
- Find the power absorbed by the resistor.
- Reverse the direction of the current source and repeat parts (a) and (b).

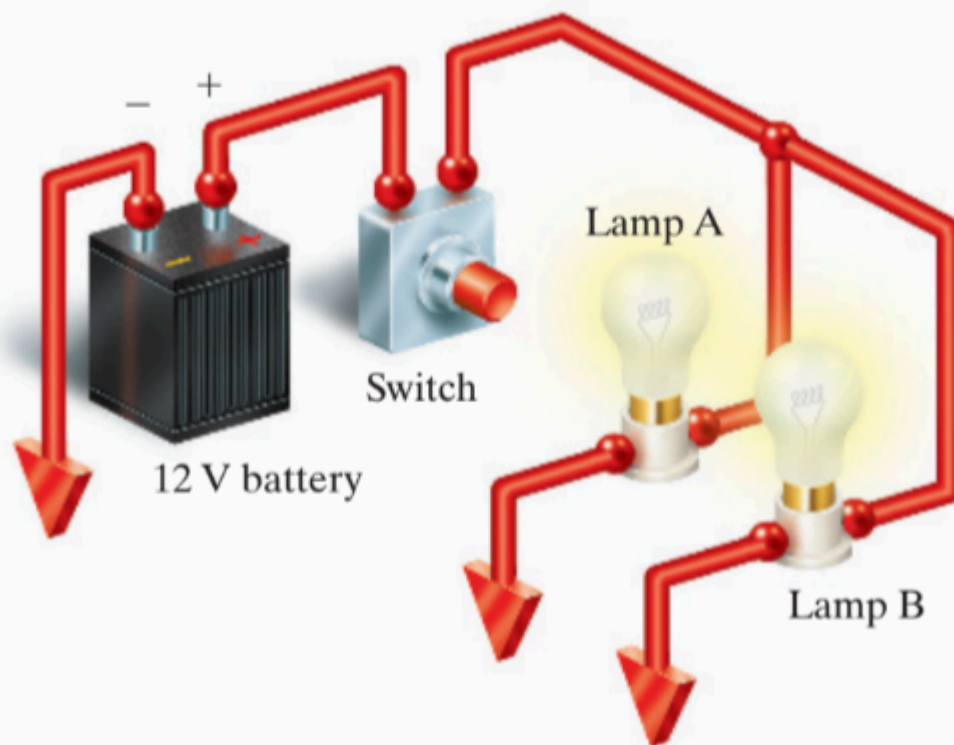
Figure P2.12



2.13 A pair of automotive headlamps is connected to a 12 V battery via the arrangement shown in Fig. P2.13. In the figure, the triangular symbol ▼ is used to indicate that the terminal is connected directly to the metal frame of the car.

- Construct a circuit model using resistors and an independent voltage source.
- Identify the correspondence between the ideal circuit element and the symbol component that it represents.

Figure P2.13

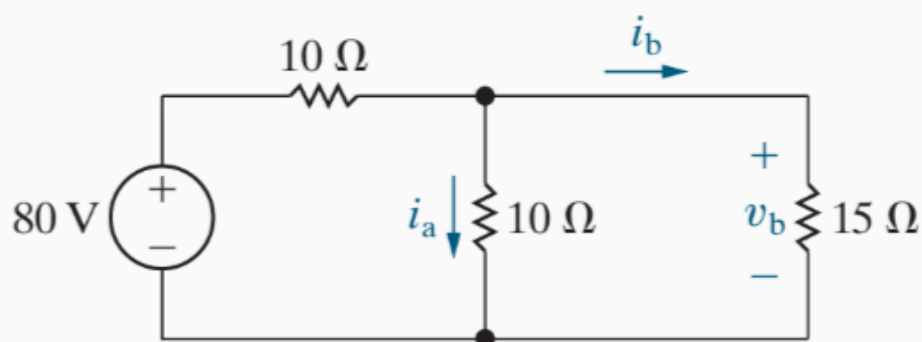


2.17 Given the circuit shown in Fig. P2.17, find

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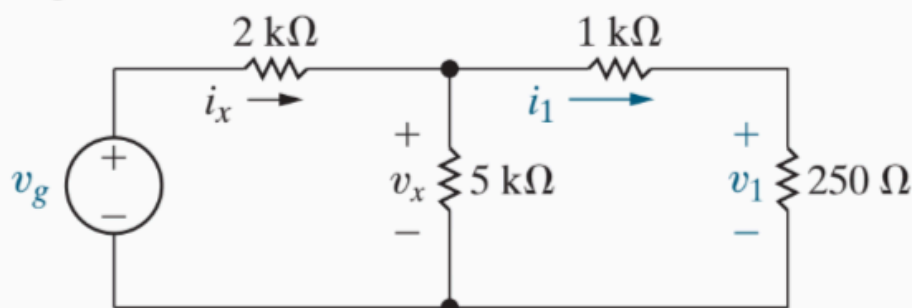
- a) the value of i_a ,
- b) the value of i_b ,
- c) the value of v_b ,
- d) the power dissipated in each resistor,
- e) the power delivered by the 80 V source.

Figure P2.17



2.18 The current i_x in the circuit shown in Fig. P2.18 is 25 mA, and the voltage v_x is 25 V. Find (a) i_1 ; (b) v_1 ; (c) v_g ; and (d) the power supplied by the voltage source.

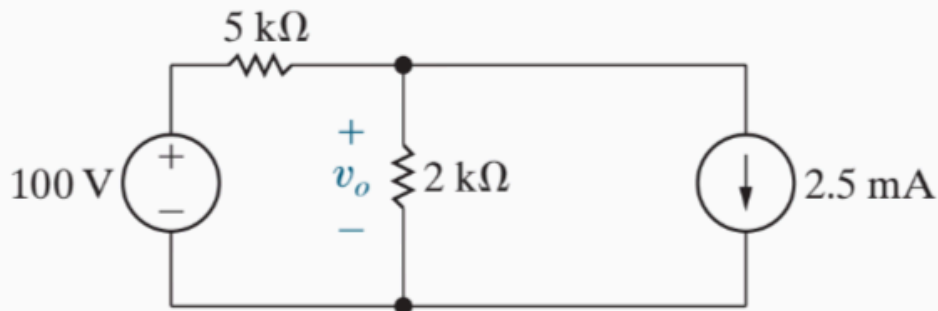
Figure P2.18



2.21 Consider the circuit shown in Fig. P2.21.

- Find v_o using Kirchhoff's laws and Ohm's law.
- Test the solution for v_o by verifying that the total power supplied equals the total power absorbed.

Figure P2.21

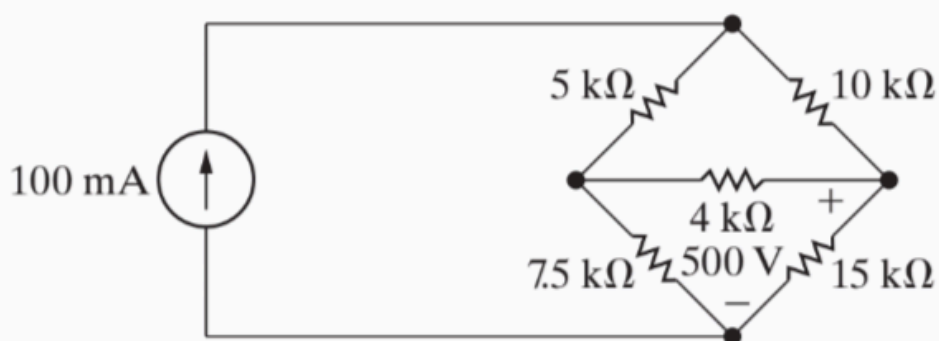


2.22 The voltage across the 15 kΩ resistor in the circuit in Fig. P2.22 is 500 V, positive at the upper terminal.

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- Find the power dissipated in each resistor.
- Find the power supplied by the 100 mA ideal current source.
- Verify that the power supplied equals the total power dissipated.

Figure P2.22

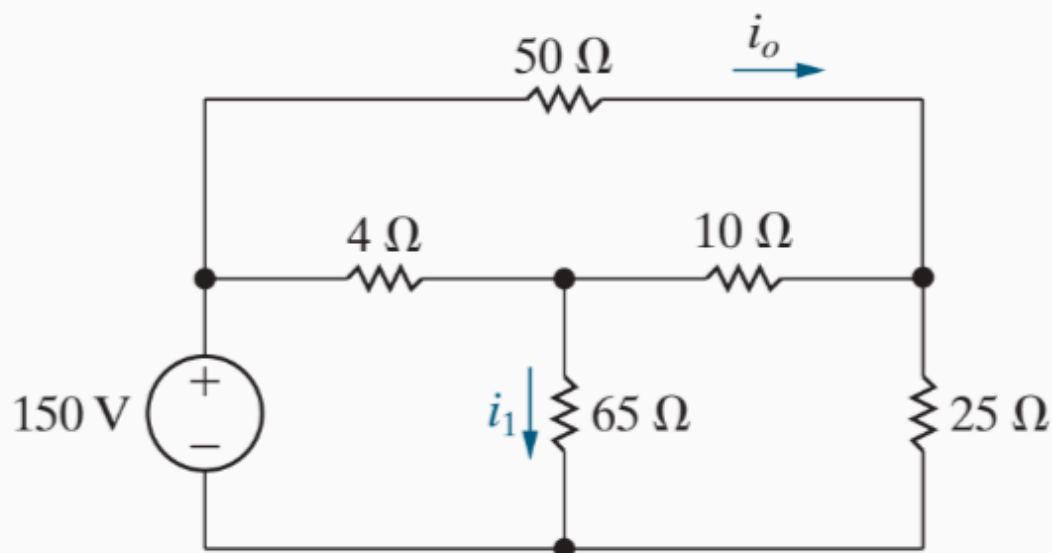


2.23 The current i_o in the circuit in Fig. P2.23 is 1 A.

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- Find i_1 .
- Find the power dissipated in each resistor.
- Verify that the total power dissipated in the circuit equals the power developed by the 150 V source.

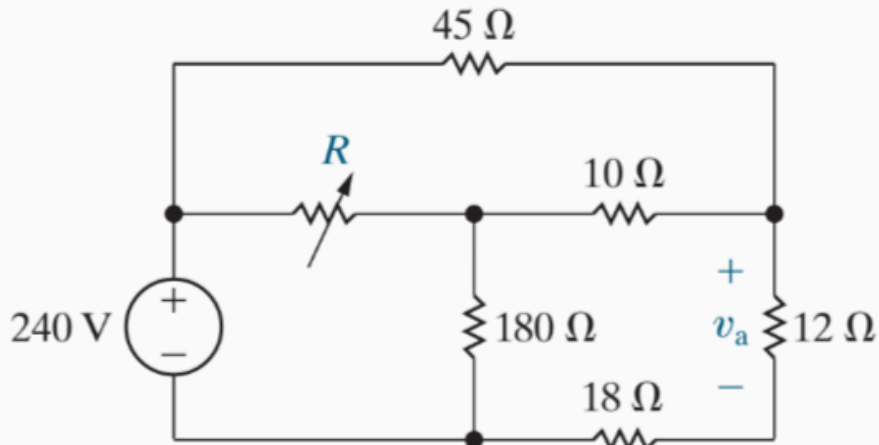
Figure P2.23



2.26 The variable resistor R in the circuit in Fig. P2.26 is adjusted until v_a equals 60 V. Find the value of R .

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Figure P2.26

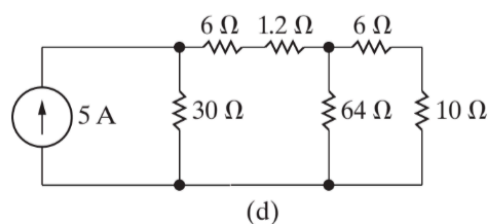
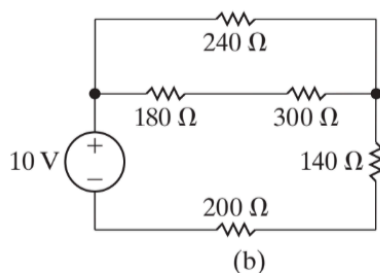
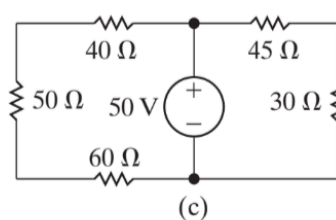
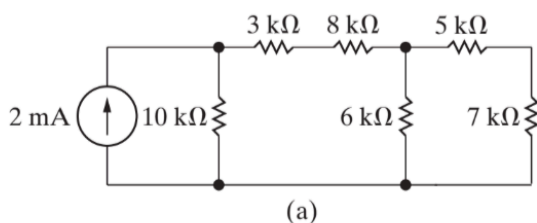


Sections 3.1–3.2

- 3.1** For each of the circuits shown in Fig. P3.1,
 a) identify the resistors connected in series,
 b) simplify the circuit by replacing the series-connected resistors with equivalent resistors.
- 3.2** For each of the circuits shown in Fig. P3.2,
 a) identify the resistors connected in parallel,
 b) simplify the circuit by replacing the parallel-connected resistors with equivalent resistors.

- 3.3** For each of the circuits shown in Fig. P3.1,
 a) find the equivalent resistance seen by the source,
 b) find the power developed by the source.
- 3.4** For each of the circuits shown in Fig. P3.2,
 a) find the equivalent resistance seen by the source,
 b) find the power developed by the source.

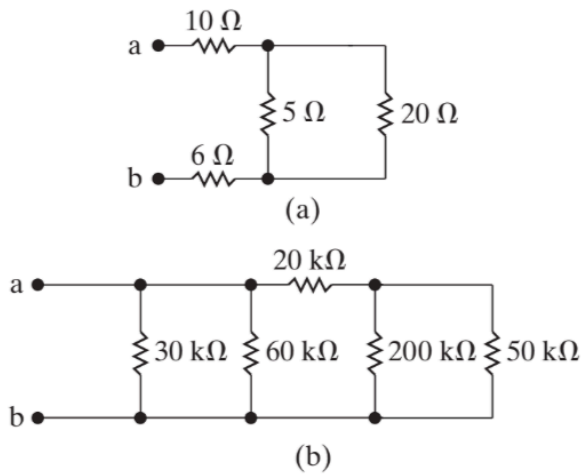
Figure P3.1



3.5 Find the equivalent resistance R_{ab} for each of the circuits in Fig. P3.5.

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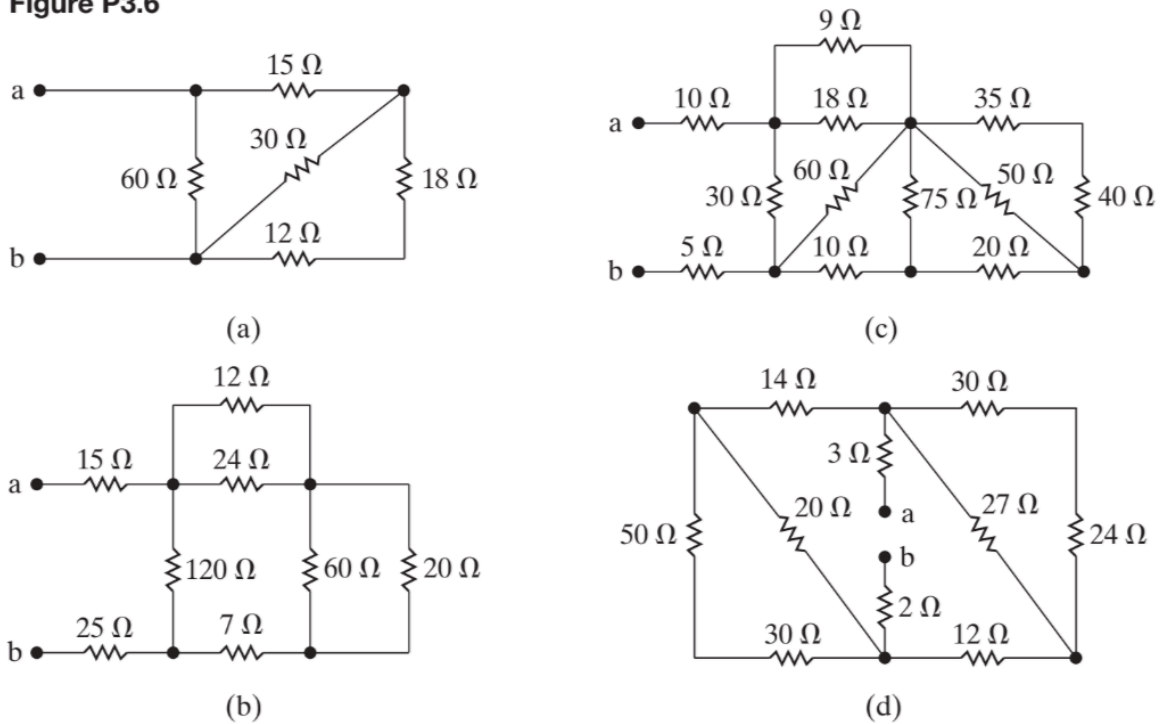
Figure P3.5



3.6 Find the equivalent resistance R_{ab} for each of the circuits in Fig. P3.6.

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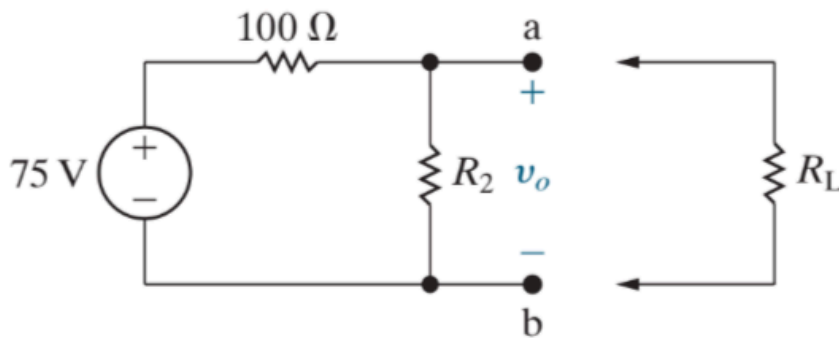
Figure P3.6



- 3.12** In the voltage-divider circuit shown in Fig. P3.12, the no-load value of v_o is 25 V. When the load resistance R_L is attached across the terminals a and b, v_o drops to 15 V. Find R_L .

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Figure P3.12



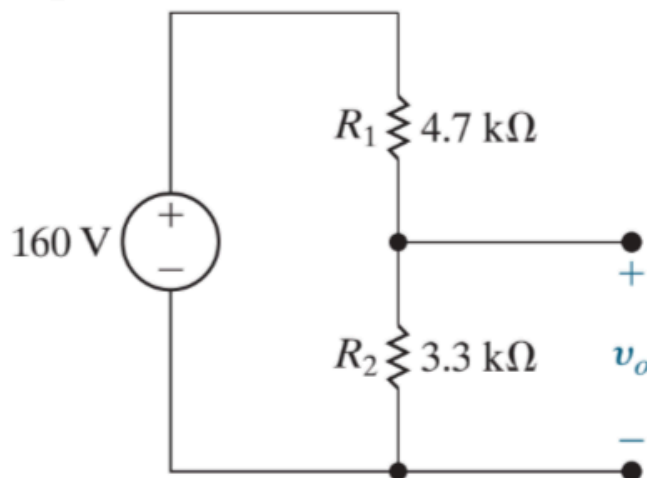
- 3.13** a) Calculate the no-load voltage v_o for the voltage-divider circuit shown in Fig. P3.13.

DESIGN
PROBLEM

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- b) Calculate the power dissipated in R_1 and R_2 .
- c) Assume that only 0.5 W resistors are available. The no-load voltage is to be the same as in (a). Specify the smallest ohmic values of R_1 and R_2 .

Figure P3.13

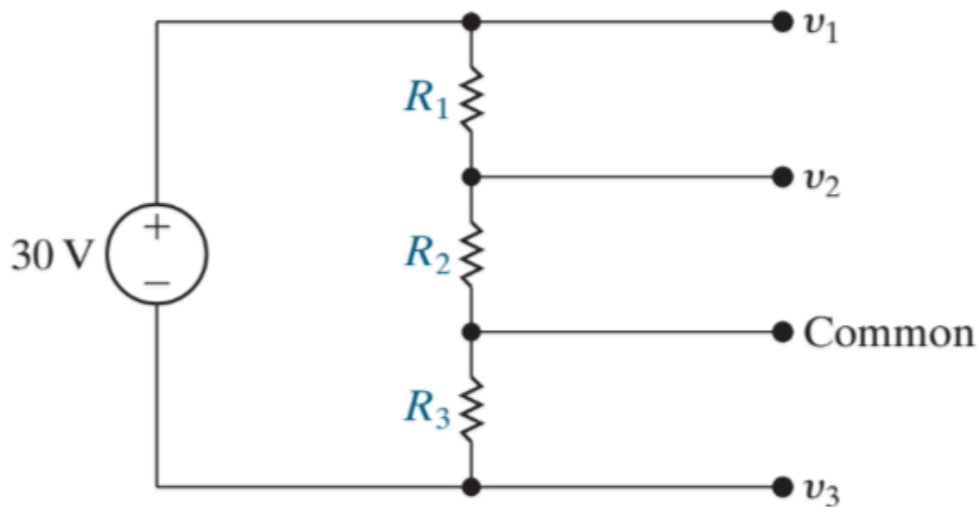


3.18**DESIGN
PROBLEM**

There is often a need to produce more than one voltage using a voltage divider. For example, the memory components of many personal computers require voltages of -15 V , 5 V , and $+15\text{ V}$, all with respect to a common reference terminal. Select the values of R_1 , R_2 , and R_3 in the circuit in Fig. P3.18 to meet the following design requirements:

- The total power supplied to the divider circuit by the 30 V source is 30 W when the divider is unloaded.
- The three voltages, all measured with respect to the common reference terminal, are $v_1 = 15\text{ V}$, $v_2 = 5\text{ V}$, and $v_3 = -15\text{ V}$.

Figure P3.18



3.20 For the current-divider circuit in Fig. P3.20 calculate

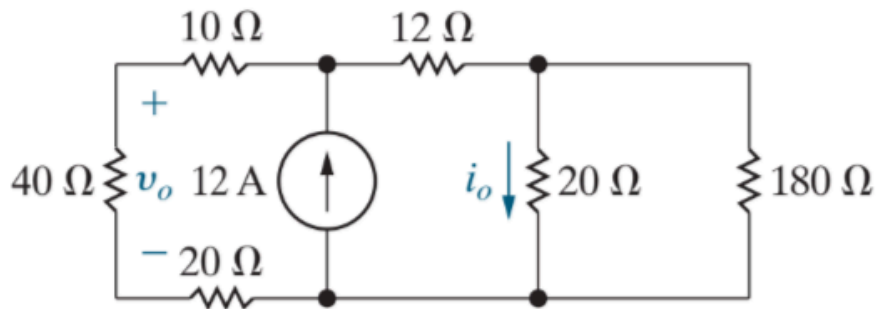
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a) i_o and v_o .

b) the power dissipated in the $12\ \Omega$ resistor.

c) the power developed by the current source.

Figure P3.20



3.52 The bridge circuit shown in Fig. 3.28 is energized from a 24 V dc source. The bridge is balanced when $R_1 = 500\ \Omega$, $R_2 = 1000\ \Omega$, and $R_3 = 750\ \Omega$.

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a) What is the value of R_x ?

b) How much current (in milliamperes) does the dc source supply?

c) Which resistor in the circuit absorbs the most power? How much power does it absorb?

d) Which resistor absorbs the least power? How much power does it absorb?

4.14 Use the node-voltage method to find the total power dissipated in the circuit in Fig. P4.14.

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Figure P4.14

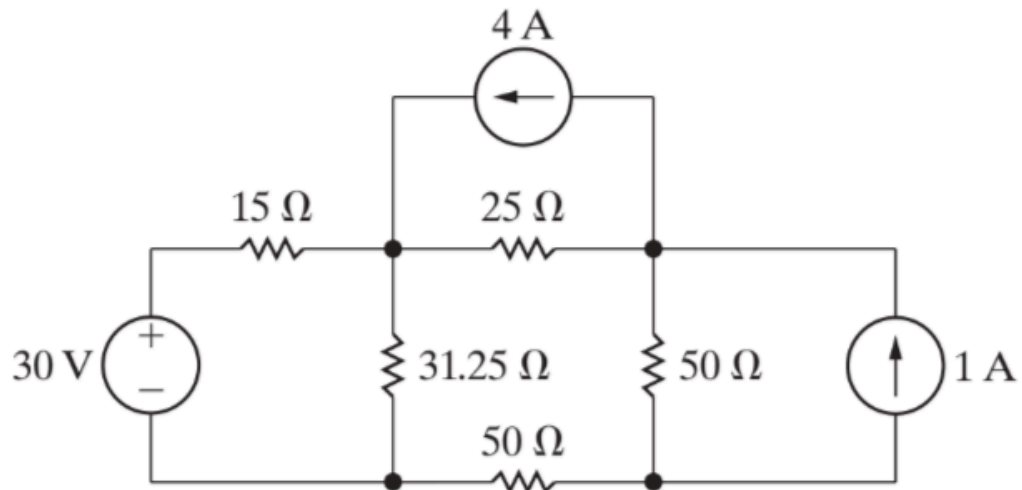


Figure P3.28

